

Mask Focal Modulation Network for Gastric Intestinal Metaplasia Segmentation

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Abstract—Gastric intestinal metaplasia (IM) is an important precancerous lesion with a high correlation to gastric cancer. Early gastric IM diagnosis is very important in reducing the risk of gastric cancer. In this paper, we propose the mask focal modulation network to achieve gastric IM segmentation from endoscopic images. The proposed network consists of the encoder module, pixel decoder module, and mask focal modulation decoder module. With the cooperation of the encoder module and pixel decoder module, the proposed mask focal modulation decoder module in the network aims to aggregate features of different field-of-views to make the network focus on the important views with learnable masks and then provide better gastric IM segmentation results. The experimental results show the superiority of the proposed method over state-of-the-art semantic segmentation methods in the gastric IM segmentation dataset. The source code is available at <https://github.com/nchucvml/MFMNetwork>.

Index Terms—Mask focal modulation network, Intestinal metaplasia, Endoscopic image, Deep learning, Semantic segmentation, Computer-aided diagnosis.

I. INTRODUCTION

Gastric intestinal metaplasia (IM) has been known a precancerous gastric lesion [1], [2] where the normal gastric mucosa is replaced by the intestinal mucosa. The severity of gastric IM is associated with the risk of gastric cancer. Statistical evidence reveals a substantial threat, with a pooled incidence rate of 12.4 per 10,000 person-years for gastric IM progressing to gastric cancer according to the American Gastroenterological Association Technical Review report [1]. To mitigate the risk of gastric cancer, early detection of gastric IM becomes an important issue.

Narrow band imaging (NBI) endoscopic images have been shown an effective tool to diagnose gastric IM without invasive biopsies [3]–[5]. However, the variations of gastric mucosa pose a significant challenge for endoscopists to diagnose gastric IM [5]–[7]. Illuminations and viewing directions of

the cameras also make the diagnosis of gastric IM become more difficult. Moreover, the diagnosis heavily relies on the experience of the endoscopists.

Recent artificial intelligence (AI) methods have shown their effectiveness for computer-aided diagnosis in different medical domains [8]–[10]. The progress of AI methods also targets diagnosing gastric IM from endoscopic images [5], [11]. These methods can be grouped into convolutional neural network (CNN)-based [11] methods and attention network-based methods [5]. CNN-based methods stack convolutional layers to make the models have better abilities to extract deep features to represent gastric IM regions. However, they are hard to capture the appearance changes under different scales or viewing directions of cameras because their receptive fields are limited by the kernel sizes. In contrast, attention-based methods are designed with specific attention modules to make the networks focus on gastric IM regions. Nevertheless, how to ensure the attention modules to specifically focus on variant gastric IM regions is still unsolved.

To solve the gastric IM segmentation problem from endoscopic images captured under different scales or viewing directions of cameras, we propose the mask focal modulation network as shown in Fig. 1. The proposed network consists of the encoder module, pixel decoder module, and mask focal modulation decoder module. The encoder is composed of focal modulation blocks [12] to capture deep features of patches of the input image under variant receptive fields. The pixel decoder [13] obtains upsampled per-pixel embeddings from the encoder features to guide the learning of the mask focal modulation decoder under different resolutions.

The mask focal modulation decoder module learns masks and prototypes with the focal modulation block to guide the network to focus on the important gastric IM regions of endoscopic images. Class embeddings and mask embeddings are obtained by the mask focal modulation decoder module to predict the class of each pixel. The losses including the binary

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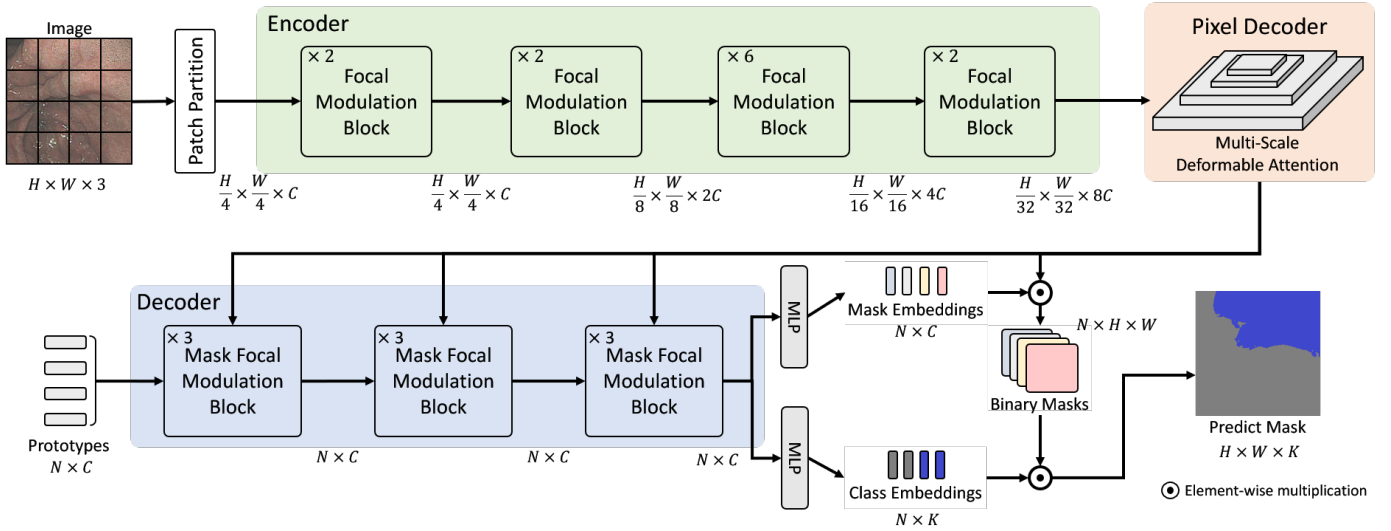


Fig. 1: The overview of the proposed mask focal modulation network.

cross-entropy loss, dice loss and classification loss are used to drive the training of the network. We conduct evaluations on the gastric IM segmentation dataset [5] and show that the proposed method is superior to state-of-the-art methods.

Our contributions can be summarized as follows:

- We propose the mask focal modulation block to aggregate features of different field-of-views with learnable masks to learn important features to represent gastric IM regions of endoscopic images.
- A novel mask focal modulation network is built by combining focal modulation encoder and pixel decoder with the mask focal modulation decoder module to achieve effective semantic segmentation which is different from the current CNN-based, attention-based, and transformer-based semantic segmentation methods.
- The proposed method achieves the best results compared with the state-of-the-art methods in the gastric IM segmentation dataset.

The paper is organized as follows. Sec. II gives the related work of semantic segmentation methods in medical images and natural images. The proposed mask focal modulation network is described in Sec. III. Experimental results with respect to state-of-the-art CNN-based methods, attention-based methods and transformer-based methods are shown in Sec. IV. Finally, Sec. V gives the conclusions.

II. RELATED WORK

A. Semantic Segmentation

Recently, CNN-based semantic segmentation methods have been shown their effectiveness in semantic segmentation of medical images. For example, U-Net [14] establishes convolutional layer based encoder-decoder structure for semantic segmentation. U-Net++ [15] considers that the model would be easier to learn when the feature maps from the decoder and encoder are semantically similar. Thus it proposes skip

pathways to connect the layers of the encoder and decoder of similar semantic feature maps. Deeplabv3+ [16] utilizes the multi-scale features in the encoder, and merges low-level features and high-level features to improve the encoder-decoder structure.

Because the field-of-views of the learned features of CNN-based methods are limited by the kernel sizes, specific attention modules are proposed to guide the model learning target regions [17], [18]. CCNet [17] designs a criss-cross attention module to capture whole image contextual information based on cross attention with the recurrent structure. SegNext [18] uses the convolution-attention method to capture larger receptive fields and encode contextual information for further improvement of semantic segmentation.

Recently, transformer [19] has been proven for its effectiveness on semantic segmentation by using the self-attention scheme for modeling long-range dependencies in the images. For example, medical transformer (MedT) [20] contains a gated axial-attention model to introduce additional controls of the self-attention module for semantic segmentation of medical images. Cheng et al. [21] propose Maskformer to perform mask classification of pixels with a set of per-pixel binary masks of each class and simplify the landscape of semantic segmentation tasks. Mask2former [13] applies masked attention to obtain localized features by constraining cross-attention centered around predicted regions. Moreover, multi-scale high-resolution features are applied in the network to help segment small objects.

Besides the aforementioned methods, Yang et al. [12] propose the focal modulation network (FocalNet). The self-attention of the transformer is replaced by a focal modulation mechanism which has hierarchical contextualization based on a stack of depth-wise convolutional layers. In this way, visual contexts from small to large field-of-views can be learned with selectively gated aggregations. The learned visual contexts are combined with the input features with element-

wise modulation for semantic segmentation.

B. Semantic Segmentation for Endoscopic Images

Recently, deep learning-based semantic segmentation methods have been applied to gastric IM segmentation from endoscopic images. For example, WDeeplab [11] based on [22] is proposed to segment gastric IM regions from endoscopic images. Pornvoraphat et al. [23] propose BiSeNet to perform gastric IM segmentation with pre-processing steps of location-wise negative sampling, jigsaw augmentation, and label smoothing. Nien et al. [5] propose CCA-MFNet which improves CCNet [17] with criss-cross attention based feature fusion encoder and feature activation map guided multi-level decoder.

Although these methods consider different mechanisms for gastric IM segmentation from endoscopic images, they still have some limitations. For example, CNN-based methods are hard to model long term dependency among patches while attention-based methods may fail to extract salient features under variant scale changes of the endoscopic images. Therefore, we propose a novel mask focal modulation network to leverage different receptive fields followed by a learnable mask mechanism to make the model focus on important regions of endoscopic images captured under variant scales and viewing directions.

III. METHOD

A. Overview

As shown in Fig. 1, the whole architecture is divided into the encoder module, pixel decoder module, and mask focal modulation decoder module. The input image is split into patches. To capture deep features of these patches under variant receptive fields, the encoder contains focal modulation blocks [12] by using a stack of depth-wise convolutional layers from small to large receptive fields. A pixel decoder [13] is used to gradually upsample low-resolution features from the output of the encoder to generate high-resolution per-pixel embeddings.

The feature embeddings of the pixel decoder are combined with the learnable prototypes by using the proposed mask focal modulation blocks in the mask focal modulation decoder. The mask focal modulation block contains learnable masks with the focal modulation blocks to guide the model to focus on the important regions for gastric IM segmentation. The mask focal modulation decoder generates class embeddings and mask embeddings after projecting the output of the decoder by using fully-connected layers. The mask embeddings are multiplied with the features of the largest resolution of the pixel decoder to generate binary masks. The binary masks are multiplied with the class embeddings to predict the class of each pixel.

The total loss ℓ is composed of three losses as follows:

$$\ell = \lambda_{ce}\ell_{ce} + \lambda_{dice}\ell_{dice} + \lambda_{cls}\ell_{cls}, \quad (1)$$

where ℓ_{ce} , ℓ_{dice} and ℓ_{cls} are the binary cross-entropy loss, dice loss, and classification loss.

For efficiency, we only calculate the cross entropy loss and dice loss for random K sample points [13]. We will give the detailed descriptions of the network in the following sections.

B. Encoder Module

Given a gastric IM segmentation dataset containing K classes, each image $x \in \mathcal{R}^{H \times W \times 3}$ is split to $M \in \mathcal{R}^{\frac{H}{4} \times \frac{W}{4}}$ patches where $W = 224$ and $H = 224$ are the image width and height, and is inputted to the proposed network. The patches are projected by a learnable linear layer to obtain the features with channels $C = 96$. The encoder applies the tiny and large receptive fields (LRF) version FocalNet [12] in the network. The encoder has four stages. The width and height of the feature maps are divided by 2 and the number of channels is doubled in each stage as shown in Fig. 1. Each stage of the encoder contains 2, 2, 6, 2 focal modulation blocks of different resolutions. With the focal modulation blocks, the network can learn visual contexts from small to large field-of-views compared with transformer-based methods.

C. Pixel Decoder Module

The pixel decoder [13] has three stages. Each stage is built by multi-scale deformable attention transformer layers [24] which has been shown powerful decoding ability from the context of different scales. The sizes of the feature embeddings of the first, second, and third stages are $1/32$, $1/16$, and $1/8$ of the original resolution in the pixel decoder. These feature embeddings are generated by using a projection network with upsampling from the previous stage. Let f_1 , f_2 , and f_3 be the output of the first, second, and third stages, respectively. The output f_4 of the pixel decoder is the feature embedding of size $1/4$ of the original resolution.

D. Mask Focal Modulation Decoder Module

The proposed mask focal modulation decoder module is shown in Fig. 1. Each stage is composed of 3 mask focal modulation blocks. There are $N = 256$ learnable prototypes served as the input of the decoder module. For each stage, prototypes $p_b \in \mathcal{R}^{N \times C}$ at block b and feature embedding f_s at stage $s \in \{1, 2, 3\}$ of the pixel encoder are used to generate the refined prototypes p_{b+1} for the following stage of the mask focal modulation decoder module.

The structure of the mask focal modulation block is shown in Fig. 2. It combines the learnable mask [13] with the focal modulation block [12]. While the focal modulation module provides information of semantic contexts of different scales, learnable masks provide segmentation specific attention to guide the model learning. Thus, we leverage the strength of these two methods.

Compared with conventional transformers, we replace the self-attention layer by the proposed mask focal modulation layer in the decoder to construct the mask focal modulation block. Specifically, two mask focal modulation layers and one feed forward network with the shortcut followed by normalization are proposed. At the b th block, each mask focal

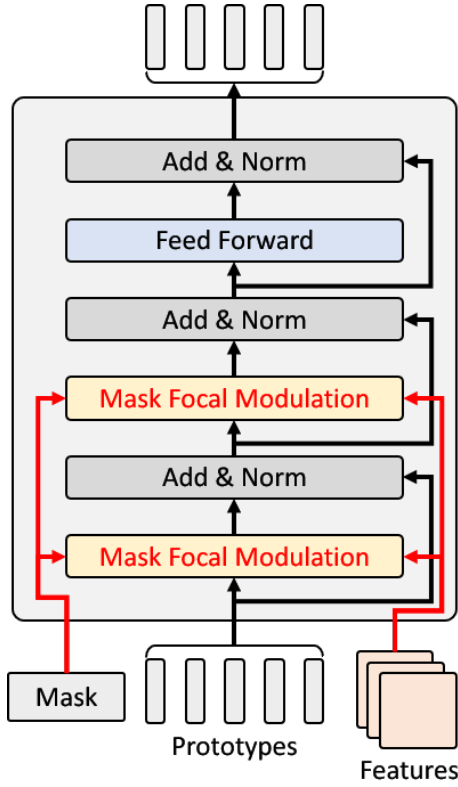


Fig. 2: The mask focal modulation block.

modulation block processes the mask $\mathbf{m}_b$, and the feature embeddings $\mathbf{f}_s$ with the prototypes $\mathbf{p}_b$.

Fig. 3 illustrates the structure of the mask focal modulation layer $MFM(\cdot)$ of the mask focal modulation block. The mask focal modulation layer uses the feature embeddings $\mathbf{f}_s$ of the pixel decoder to guide the learning of the prototypes $\mathbf{p}_b$ and utilizes the learnable mask $\mathbf{m}_b$ to make the network focus on learning important regions at stage s and block b . The $MFM(\cdot)$ in each stage can be formulated as follows:

$$MFM(\mathbf{p}_b, \mathbf{f}_s, \mathbf{m}_b) = Z(\mathbf{p}_b) \odot F(R(\mathbf{f}_s)) \odot \mathbf{m}_b, \quad (2)$$

where $Z(\cdot)$ is the focal modulation function, $F(\cdot)$ indicates the projection function, $R(\cdot)$ represents the resize function, and $\odot$ represents the element-wise multiplication. Here, $\mathbf{m}_b$ is a binary mask at block b . It is the resized mask prediction $\mathcal{M}$ from the previous block $b-1$ with the threshold of 0.5. The pixel value of mask $\mathbf{m}_b$ at the location (x, y) can be denoted as follows:

$$\mathbf{m}_b(x, y) = \begin{cases} 1, & \text{if } \mathcal{M}_{b-1}(x, y) < 0.5 \\ 0, & \text{otherwise} \end{cases}. \quad (3)$$

In this way, we can enforce the network to learn the pixels that are hard to be classified by using the mask $\mathbf{m}_b$.

The focal modulation function $Z(\cdot)$ can be divided into gate mechanism $G^l(\cdot)$ and context aggregation $A^l(\cdot)$ [12]. To

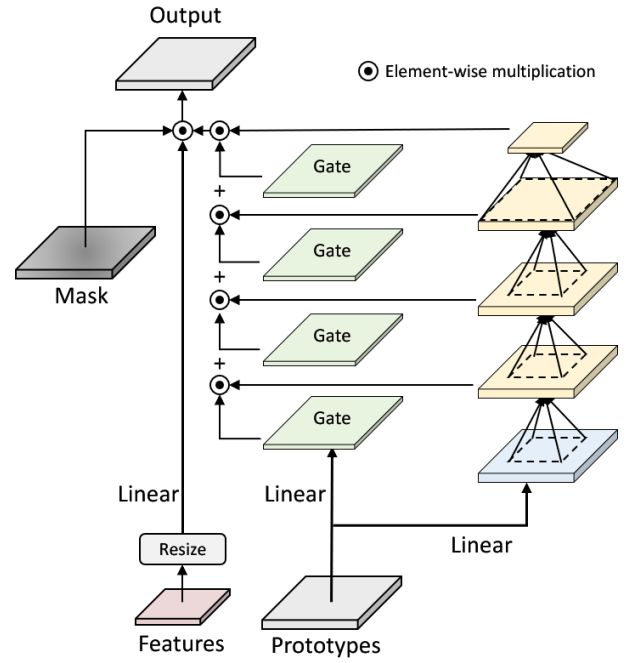


Fig. 3: The mask focal modulation layer.

utilize the features from different receptive fields, the focal modulation $Z(\cdot)$ is represented as follows:

$$Z(\mathbf{p}_b) = \sum_{l=1}^{L+1} G^l(\mathbf{p}_b) \odot A^l(\mathbf{p}_b), \quad (4)$$

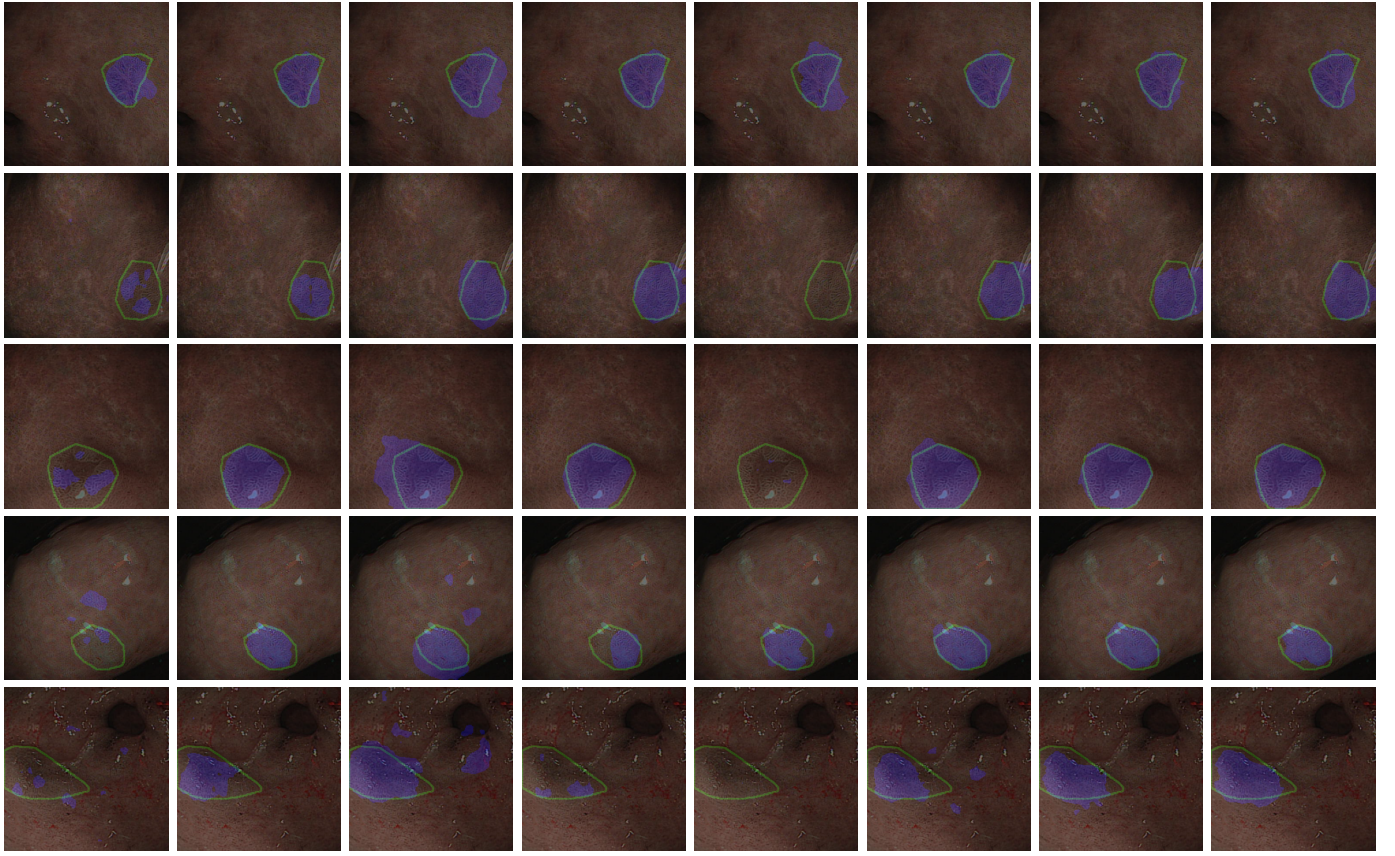
where $l \in \{1, \dots, L\}$ indicates the index of the focal level and $L = 3$. The focal modulation is controlled by the gate mechanism $G^l(\cdot)$. $G^l(\cdot)$ is a slice channel of $G(\mathbf{p}_b) = f_g(\mathbf{p}_b)$ at focal level l after the prototypes are projected with a linear layer $f_g(\cdot)$.

The other important module in the focal modulation [12] is context aggregation. Context aggregation $A^l(\cdot)$ aggregates the contexts from the previous layer $A^{l-1}(\cdot)$ by using the depth-wise convolution with the kernel size of 9 followed by the $GELU(\cdot)$ activation function. $A^l(\mathbf{p}_i)$ is formulated as follows:

$$A^l(\mathbf{p}_b) = GELU(DW(A^{l-1}(\mathbf{p}_b))), \quad (5)$$

where the first layer of the context aggregation A^0 is the prototypes after projection $A^0(\mathbf{p}_b) = f_g(\mathbf{p}_b)$, and $DW(\cdot)$ is the depth-wise convolution function. For the last focal level, the context aggregation $A^L(\mathbf{p}_b) = AP(A^L(\mathbf{p}_b))$, where $AP(\cdot)$ is the average pooling layer. In this way, the context aggregation is achieved by a stack of depth-wise convolutions. Moreover, the learned features contain the information of different receptive fields which help learn effective features to represent the gastric IM regions captured under variant scales and viewing angles.

After the mask focal modulation decoder module, class embeddings and mask embeddings are obtained by projecting the output of the mask focal modulation decoder module with



(a) U-Net (b) W-Deeplab (c) CCA-MFNet (d) SegNext (e) MedT (f) Mask2Former (g) FocalNet (h) Proposed

Fig. 4: Qualitative results of the proposed method and the state-of-the-art methods with respect to gastric IM regions in the endoscopic images. (a) U-Net, (b) W-Deeplab, (c) CCA-MFNet, (d) SegNext, (e) MedT, (f) Mask2Former, (g) FocalNet, and (h) Proposed method.

fully-connected layers. The mask embeddings are multiplied with the features of the largest resolution of the pixel decoder to generate binary masks which help the network learn pixels which are hard to be learned. The class embeddings are multiplied with the binary masks to obtain the segmentation of each pixel. Then, the loss ℓ shown in Eq. (1) is computed as shown in [13] to drive the training of the network.

IV. EXPERIMENTAL RESULTS

A. Experimental Settings

Dataset. We evaluated our method on the gastric IM segmentation dataset [5], containing various gastric sections of NBI images. The dataset includes 317 (70%) training images and 137 (30%) testing images. Data augmentation was applied including image brightness, image contrast, random crops, flips, color jitters, and rotations [25]. The augmentation expands our training dataset to 12,363 images to mitigate the risk of overfitting.

We used mean intersection-over-Union (mIoU) [26] and mean dice (mDice) [27] to evaluate the performance of the proposed method and competing methods. Besides, we also

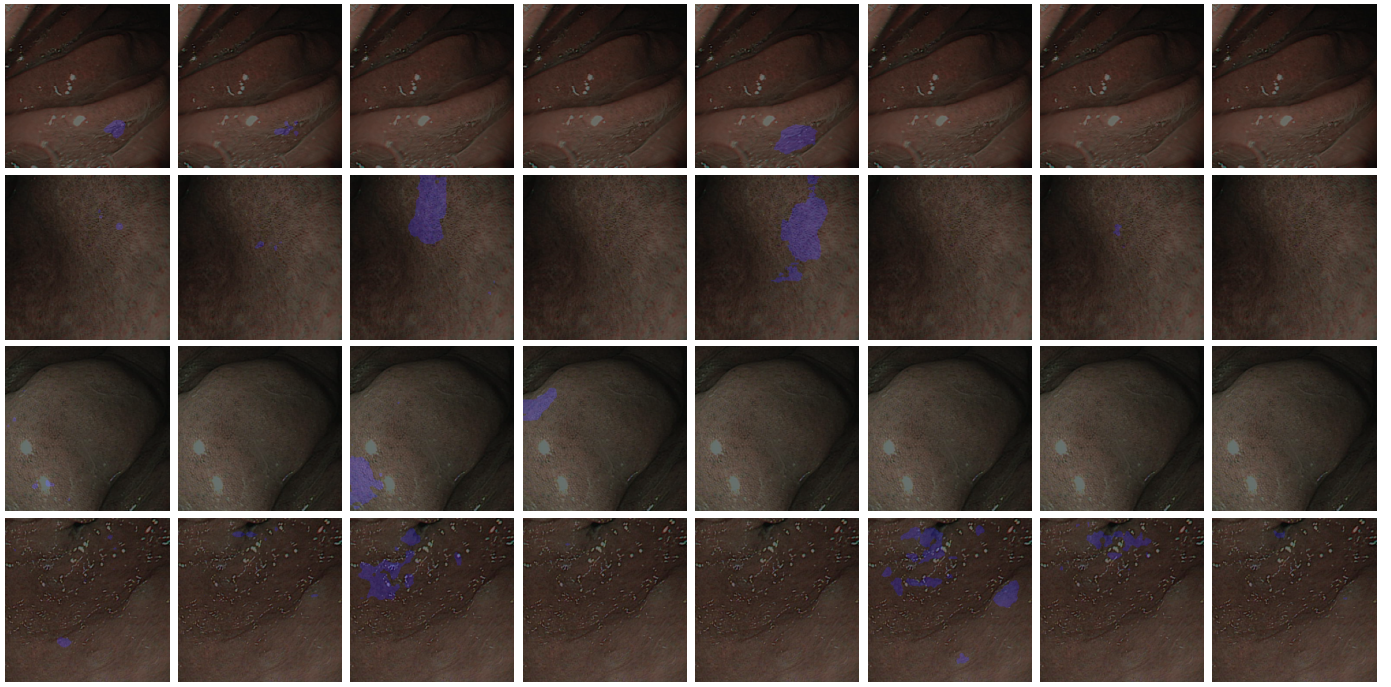
used mean recall (mRecall) and mean precision (mPrecision) for evaluations.

Settings. We implemented the proposed method based on [28]. The dataset was run on a GTX 3080 GPU and all images were resized to 224×224 . We used the AdamW optimizer [29] with a learning rate starting at 6×10^{-5} and weight decay of 0.01. The training epochs were set as 50 with a batch size of 16. The values of λ_{ce} , λ_{dice} , and λ_{cls} were empirically set as 5, 5, and 2, respectively.

We compared the proposed method with state-of-the-art CNN-based methods including U-Net [14] and W-Deeplab [11], attention-based methods including CCA-MFNet [5] and SegNext [18], transformer-based methods including MedT [20] and Mask2former [13], and FocalNet [12], where the results of U-Net, W-Deeplab, MedT, and CCA-MFNet were reported in [5].

B. Qualitative Results

Fig. 4 and Fig. 5 show the qualitative results of the state-of-the-art methods and the proposed method with respect to gastric IM regions and normal regions of endoscopic images. The ground truths of gastric IM regions are marked by light



(a) U-Net (b) W-Deeplab (c) CCA-MFNet (d) SegNext (e) MedT (f) Mask2Former (g) FocalNet (h) Proposed

Fig. 5: Qualitative results of the proposed method and the state-of-the-art methods with respect to normal regions in the endoscopic images. (a) U-Net, (b) W-Deeplab, (c) CCA-MFNet, (d) SegNext, (e) MedT, (f) Mask2Former, (g) FocalNet, and (h) Proposed method.

green polygons and the predicted gastric IM regions of each model are marked by purple colors.

Fig. 4(a), (b), (c), (d), (e), (f), (g), and (h) show the segmentation results of gastric IM regions by using U-Net, W-Deeplab, CCA-MFNet, SegNext, MedT, Mask2Former, FocalNet, and the proposed method, respectively. While U-Net fails to find the gastric IM regions, W-Deeplab achieves better results due to its design for gastric IM segmentation. Such results show that the task-specific design for gastric IM segmentation is important.

Although attention-based methods achieve better mIoU values compared with CNN-based methods, false alarms are shown in Fig. 4(c) of CCA-MFNet, and miss-segmentation results are shown in Fig. 4(d) of SegNext. As shown in Fig. 4(e), the transformer-based method, MedT, achieves worse results because of lacking sufficient training data. In contrast, Mask2Former shown in Fig. 4(f) achieves better results by considering features of different scales to increase the feature representation ability under a small training set. Compared with the above methods, FocalNet achieves comparable results compared with Mask2Former as shown in Fig. 4(g). By considering mask focal modulation blocks, the proposed method obtains the best results for gastric IM segmentation as shown in Fig. 4(h).

Fig. 5(a), (b), (c), (d), (e), (f), (g), and (h) show the segmentation results of normal regions by using U-Net, W-Deeplab, CCA-MFNet, SegNext, MedT, Mask2Former, FocalNet, and the proposed method, respectively. Compared with CNN-based

methods shown in Fig. 5(a) and (b), attention-based methods have more false alarms because special designed attention schemes are used to force the network to focus on specific regions as shown in Fig. 5(c) and (d). Such situations can also be observed in Fig. 5(e) and (f) of transformer-based methods. The results of FocalNet shown in Fig. 5(g) also contain false alarms. In contrast, by inducing the mask scheme, the proposed method can achieve better results. In summary, with the proposed mask focal modulation block, the proposed method can learn better feature representations of the gastric IM regions and distinguish normal regions from gastric IM regions more effectively.

C. Quantitative Results

Table I presents the quantitative results including mIoU, mDice, mRecall and mPrecision of the proposed method and the state-of-the-art methods. U-Net only has 57.61% mIoU because its architecture is hard to distinguish the gastric IM regions from normal regions under a small training set. In contrast, W-Deeplab is specifically designed for gastric IM segmentation, and thus it achieves significant better results. Compared with CNN-based methods, attention-based methods achieve better mIoU results because their attention schemes can help the network learn key features. However, their mPrecision results are worse than those of W-Deeplab. Transformer-based methods such as Mask2former consider the self-attention scheme which generates global views of the endoscopic images. Thus, Mask2former achieves even

TABLE I: Comparisons with state-of-the-art methods.

Type	Model	Backbone	mIoU	mDice	mRecall	mPrecision
Convolution-based	U-Net [14]	U-Net	57.61	65.60	61.40	77.18
	W-Deeplab [11]	ResNet-50	65.85	75.26	69.09	89.14
Attention-based	CCA-MFNet [5]	ResNet-50	68.92	78.47	74.94	83.45
	SegNext [18]	Mscan-s	68.87	78.36	73.07	87.55
Transformer-based	MedT [20]	Transformer	53.31	59.44	56.62	71.03
	Mask2former [13]	Swin-tiny	71.32	80.71	77.16	85.53
Focal modulation-based	FocalNet [12]	FocalNet-tiny	70.98	80.44	78.27	83.04
	Proposed	FocalNet-tiny	73.33	82.49	79.33	86.56

better mIoU results compared with attention-based methods. In contrast, MedT performs worse because of lacking sufficient training data. By considering the mask scheme with the focal modulation of different receptive fields, the proposed method achieves the best results and has around 2.3% and 2.1% improvement of mIoU and mDice compared with FocalNet. Such results demonstrate the effectiveness of the proposed mask focal modulation block for gastric IM segmentation under variant appearance changes and viewing directions of the endoscopic cameras.

V. CONCLUSIONS

We propose the mask focal modulation network to solve the gastric IM semantic segmentation problem. By imposing the learnable masks to the focal modulation block, the proposed network is able to learn more accurate gastric IM boundaries and outperforms state-of-the-art methods. In the future, we will apply the proposed network to different medical images including pathology images [30], [31], and CT images [8], [32] to evaluate the generalization ability.

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